Supplementary Material for "Affordable Real Style Transfer: Training Spectral Flow Matching on an RTX 3060 in Minutes"

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A Method Details

This section expands Section 3 of the main paper. All theorem, lemma, proposition, and definition numbers refer to the main paper.

A.1 Problem Setup

Let $x_c \in \mathbb{R}^{H\times W\times 3}$ be a content image and $s \in \{1,\ldots,K\}$ a target style domain. We learn a mapping $G_\theta(x_c, s) = \mathcal{D}(\hat{z}_s)$, where $\mathcal{D}$ is a frozen VAE decoder and $\hat{z}_s$ is a stylized latent. The transport is a deterministic ODE (rectified flow) from $z_0 = z_c$ to $z_1 \sim p_{\text{style}}$.

VAE latent choice. We work in the latent space of a pre-trained SDXL VAE, $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4\times 32\times 32}$. Three properties motivate this choice. First, the decoder is frozen and perceptually calibrated, so a small edit in latent space maps to a semantically meaningful edit in pixel space. Second, the 32×32 spatial size yields 16×16 subbands after a single-level Haar DWT, which a small backbone can process at batch 24 on a 12 GB GPU. Third, four channels match the standard SDXL latent convention, allowing reuse of off-the-shelf VAE weights without retraining.

A.2 Haar DWT and Spectral Decoupling

We restate Lemma 1 and Proposition 1 of the main paper and give full proofs.

Lemma (Wavelet isometry and Parseval identity — Lemma 1 of the main paper)**.** *The single-level 2D Haar DWT $\mathcal{W} : \mathbb{R}^{C\times H\times W} \to \mathbb{R}^{C\times H/2\times W/2} \times \{4 \text{ subbands}\}$ is an isometry: $\|\mathcal{W}(z)\|_F^2 = \|z\|_F^2$. For any partition of the four subbands into disjoint sets A, B, $\|z\|_F^2 = \|A\|_F^2 + \|B\|_F^2$ with no cross term.*

Proof. The Haar kernel is $H_2 = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right]$, satisfying $H_2^\top H_2 = I_2$. The 2D transform applies H_2 separably along rows then columns, so the per-patch transform is the Kronecker product $H_2 \otimes H_2$. Kronecker products preserve orthogonality: $(A\otimes B)^\top(A\otimes B) = (A^\top A)\otimes(B^\top B) = I_2 \otimes I_2 = I_4$. Hence $\mathcal{W}$ is an orthogonal linear map and $\|\mathcal{W}(z)\|_F^2 = \|z\|_F^2$ (Parseval). Orthogonality of the four subband blocks gives $\|z\|_F^2 = \|\text{LL}\|_F^2 + \|\text{LH}\|_F^2 + \|\text{HL}\|_F^2 + \|\text{HH}\|_F^2$, so any partition $A \cup B$ satisfies $\|z\|_F^2 = \|A\|_F^2 + \|B\|_F^2$ with no cross term. □

Proposition (Spectral content–style decoupling — Proposition 1 of the main paper)**.** *If the content signal concentrates its energy in LL, $\|z_{LL}^{cnt}\|_F^2 \geq (1-\epsilon)\|z^{cnt}\|_F^2$, then any transform T acting only on {LH,HL,HH} perturbs the content energy by at most $\epsilon\,\|z^{cnt}\|_F^2$.*

Proof. By Lemma 1, the LL and HF subspaces are orthogonal, so $\|T(z) - z\|_F^2 = \|T(z^{\text{sty}}) - z^{\text{sty}}\|_F^2 + \|T(z_{\text{HF}}^{\text{cnt}}) - z_{\text{HF}}^{\text{cnt}}\|_F^2$. Since T acts only on HF, $T(z_{\text{LL}}^{\text{cnt}}) = z_{\text{LL}}^{\text{cnt}}$, and the second term is bounded by $\|z_{\text{HF}}^{\text{cnt}}\|_F^2 \leq \epsilon\,\|z^{\text{cnt}}\|_F^2$ by the energy-concentration assumption. □

A.3 Base Locking

We set the LL velocity loss weight $w_{\text{LL}} = 0$ during training but still apply the small predicted LL drift at inference. The training loss is

$$\mathcal{L}(\theta) = \sum_{c\in\{\text{LH, HL}\}} w_c\, \mathbb{E}_t \big\| v_\theta^c(z_t, t, s) - u_t^c \big\|_2^2, \qquad w_{\text{LH}} = w_{\text{HL}} = 1,\ w_{\text{HH}} = 0.$$

Why $w_{\mathbf{LL}} = 0$. By Proposition 1, content energy concentrates in LL. Training an LL velocity would couple content-reconstruction gradients with style gradients in the same subspace, re-creating the degeneration attractor (Theorem 1) inside the LL block: the LL head would be pulled toward the style-averaged velocity $\bar{v}_\theta$, the only critical point at which content and style gradients cancel. Setting $w_{\mathrm{LL}} = 0$ removes the LL term from the flow-matching loss, so no gradient flows to v_θ^{LL} through $\mathcal{L}_{\mathrm{FM}}$. The LL head drifts only through shared backbone weights and remains small in norm.

Why LL is not frozen at inference. LL also carries global style (color palette, lighting). Freezing LL at inference removes the only path by which the model can shift global tone, costing 0.0141 CLIP-S (Table 2). Letting the small freely-drifting predicted LL velocity be applied at inference gives both the content anchor (LL is barely trained, so v_θ^{LL} is small) and a global-style channel. The two objectives — content preservation in LL and style injection in HF — no longer compete because they act on orthogonal subspaces (Lemma 1).

A.4 Fiber Flow and WCT

Definition (Spectral fiber bundle — Definition 1 of the main paper)**.** *The total space $E = \mathbb{R}^{C\times H\times W}$ is equipped with the projection $\pi : E \to B$ that returns the LL subband, $\pi(z) = z^{LL} \in B$. The fiber above $b \in B$ is $F_b = \{z : z^{LL} = b\}$, parameterized by (LH,HL,HH). A map $T : E \to E$ is* fiber-preserving *if $\pi(T(z)) = \pi(z)$ for all z.*

Proposition (WCT is fiber-preserving — Proposition 2 of the main paper)**.** *Per-subband whitening-and-coloring on each high-frequency subband leaves LL unchanged, hence $\pi(\mathrm{WCT}(\hat{z}_s)) = \pi(\hat{z}_s) = \hat{\ell}$. The content anchor is an invariant of style injection; style transfer is a vertical map along the fibers F_b.*

Proof. For each $c \in \{\mathrm{lh,hl,hh}\}$, $\mathrm{WCT}_c(\hat{c}, c^{\mathrm{ref}}) = \Sigma_{c^{\mathrm{ref}}}^{1/2}\,\Sigma_{\hat{c}}^{-1/2}\,(\hat{c} - \mu_{\hat{c}}) + \mu_{c^{\mathrm{ref}}}$ depends only on the HF subband c. LL is not an argument of any WCT_c, so $\ell^{\mathrm{new}} = \hat{\ell}$ by construction. Therefore $\pi(\mathrm{WCT}(\hat{z}_s)) = \hat{\ell} = \pi(\hat{z}_s)$, which is the fiber-preserving condition. By Lemma 1, the four subbands are orthogonal, so per-fiber statistics matching is equivalent to per-frequency-band style injection in the original latent, with no cross-band leakage. □

A.5 End-of-Trajectory Injection

Proposition (α-identifiability under endpoint injection — Proposition 3 of the main paper)**.** *Let $r_n(\alpha) = (1-\alpha)^n$ be the content residual after n per-step injections with blend $\alpha \in (0, 1]$. Under per-step injection with $n \geq 2$, the set of δ-indistinguishable pairs covers $[0, 1]^2$ for any $\delta > (1-\alpha_{\min})^n$. Under endpoint injection ($n = 1$), $r_1(\alpha) = 1-\alpha$ is injective, so all pairs are distinguishable for any $\delta < 1$.*

Proof. For $n \geq 2$, $r_n(\alpha) = (1 - \alpha)^n$ is convex and decreasing on $[0, 1]$ with $r_n(0) = 1$, $r_n(1) = 0$, and second derivative $n(n-1)(1-\alpha)^{n-2} \geq 0$. Its derivative magnitude is $|r_n'(\alpha)| = n(1-\alpha)^{n-1}$, which tends to 0 as $\alpha \to 1$: near $\alpha = 1$, r_n is flat, so a continuum of α values produce residuals within δ of each other. On $[\alpha_{\min}, 1]$ the Lipschitz constant is $n(1-\alpha_{\min})^{n-1}$; for $n = 8$, $\alpha_{\min} = 0.5$ this is $8 \cdot 0.5^7 \approx 0.0625$, so resolution below 0.0625 is lost. Hence for any $\delta > (1 - \alpha_{\min})^n$ the δ-indistinguishable pairs cover the square. For $n = 1$, $r_1(\alpha) = 1 - \alpha$ is affine with slope -1, hence strictly monotone and injective. Distinct α therefore yield distinct residuals, and any two blends are distinguishable for every $\delta < 1$. □

This is why style is injected only at the last ODE step: it restores α as a meaningful trade-off knob and matches the Schrödinger-bridge view in which style is a terminal constraint, not a per-step perturbation.

A.6 Stochastic Frequency Routing

The cross-attention that routes style tokens into the velocity field can operate on the full latent (Euclidean query) or on the high-frequency tokens only (wavelet query). At inference we always use the wavelet query. At training we draw $B \sim \mathrm{Bernoulli}(p)$ per step and use the wavelet query with probability p. The sweep $p \in \{0.0, 0.5, 0.8, 1.0\}$ gives

$$\text{CLIP-S} \in \{0.7061,\ 0.7083,\ 0.7213,\ 0.7226\}, \quad \text{LPIPS} \in \{0.2606,\ 0.2480,\ 0.2868,\ 0.3068\}.$$

Why $p = 0.8$**.** Two failure modes bracket the sweep. At $p = 0.0$ the model never sees a wavelet query during training, so the query projection is mis-specified at inference and CLIP-S drops to 0.7061 (a train/test mismatch). At $p = 1.0$ the style encoder over-specializes to high-frequency statistics and loses global-palette information, which inflates LPIPS to 0.3068 through content tearing. The point $p = 0.8$ is the unique sweep value that clears both thresholds (CLIP-S $>$ 0.72 and LPIPS $<$ 0.30). Mixing the two query modes regularizes the projection: biasing toward the inference-time (wavelet) query while retaining enough Euclidean exposure to preserve palette information.

A.7 Degeneration Attractor

Definition (Degeneration manifold — Definition 2 of the main paper)**.** $\mathcal{M}_0 = \{\theta : v_\theta(z,t,s) = \bar{v}_\theta(z,t) = \mathbb{E}_s[v_\theta(z,t,s)]\ \forall s,z,t\}$, *equivalently* $\nabla_s v_\theta|_{\theta\in\mathcal{M}_0} = 0$.

Theorem ($\mathcal{M}_0$ is a local minimum under multiplicative attenuation — Theorem 1 of the main paper)**.** *Assume* $\mathcal{L} = w_{FM}\mathcal{L}_{FM} + w_{SWD}\mathcal{L}_{SWD}$ *with* $w_{FM} \gg w_{SWD}$*, and three multiplicative attenuation factors act on the style gradient: a scalar gate* $g(\theta) \in [0,1]$*, an attention-entropy factor* $h(\theta) \in [0,1]$ *(*$h = 1$ *uniform), and a normalization washout* $n(\theta) \in [0,1]$*. Let* $\kappa(\theta) = g(\theta)\,h(\theta)\,n(\theta)$*. If at* $\theta_0 \in \mathcal{M}_0$*: (i)* $\kappa(\theta_0) \le \sqrt{w_{SWD}/w_{FM}}$*; (ii)* $\nabla^2_{\theta_\perp}\mathcal{L}_{FM}|_{\theta_0}$ *is positive definite on* $T^\perp_{\theta_0}\mathcal{M}_0$*; (iii)* $\|\nabla^2\mathcal{L}_{SWD}|_{\theta_0}\|_{op} \le \kappa(\theta_0)\,\|\nabla^2\mathcal{L}_{FM}\|_{op}$*; then* θ_0 *is a strict local minimum of* $\mathcal{L}$*, with basin of attraction of radius* $\Theta(\kappa(\theta_0))$*.*

Proof. Gradient attenuation. The scalar gate g multiplies the style-conditioned signal in the velocity field; uniform attention (entropy factor $h\to 1$) spreads style tokens evenly so their weighted sum washes out the style direction; GroupNorm (factor n) removes per-channel mean and variance, which are precisely the channels that carry style statistics. Composing these, the style gradient satisfies $\|\nabla_\theta\mathcal{L}_{\rm SWD}\| \le g\,h\,n\,\|\nabla_\theta\mathcal{L}_{\rm FM}\| = \kappa\,\|\nabla_\theta\mathcal{L}_{\rm FM}\|$.

First-order condition. On $\mathcal{M}_0$, $v_\theta = \bar{v}_\theta$ is style-independent, i.e. the conditional mean over styles. This is the minimizer of $\mathcal{L}_{\rm FM}$ averaged over s, so $\nabla_\theta\mathcal{L}_{\rm FM}|_{\theta_0} = 0$. By condition (i), $\|\nabla_\theta\mathcal{L}_{\rm SWD}\| \le \kappa\,\|\nabla_\theta\mathcal{L}_{\rm FM}\| = 0$, so $\nabla\mathcal{L}(\theta_0) = w_{\rm FM}\nabla\mathcal{L}_{\rm FM} + w_{\rm SWD}\nabla\mathcal{L}_{\rm SWD} = 0$.

Second-order condition. Decompose a perturbation as $\theta = \theta_0 + \delta\theta_\perp + \delta\theta_\parallel$ with $\delta\theta_\perp \in T^\perp_{\theta_0}\mathcal{M}_0$. The second-order expansion along the normal direction is

$$\delta^2\mathcal{L} = \tfrac{1}{2}\,\delta\theta_\perp^\top\big(\nabla^2_\perp\mathcal{L}_{\rm FM} + (w_{\rm SWD}/w_{\rm FM})\nabla^2_\perp\mathcal{L}_{\rm SWD}\big)\delta\theta_\perp.$$

By (ii), $\lambda_{\min}(\nabla^2_\perp\mathcal{L}_{\rm FM}) > 0$. By (iii), $\|\nabla^2_\perp\mathcal{L}_{\rm SWD}\|_{\rm op} \le \kappa\,\|\nabla^2\mathcal{L}_{\rm FM}\|_{\rm op}$. Therefore

$$\delta^2\mathcal{L} \ge \tfrac{1}{2}\big(\lambda_{\min}(\nabla^2_\perp\mathcal{L}_{\rm FM}) - (w_{\rm SWD}/w_{\rm FM})\,\kappa\,\|\nabla^2\mathcal{L}_{\rm FM}\|_{\rm op}\big)\|\delta\theta_\perp\|^2.$$

By (i), $(w_{\rm SWD}/w_{\rm FM})\,\kappa \le \kappa^2/\kappa \le \kappa$, so the bracket is positive whenever $\kappa < \lambda_{\min}/\|\nabla^2\mathcal{L}_{\rm FM}\|_{\rm op}$, which holds within radius $O(\kappa)$ of θ_0. Hence $\delta^2\mathcal{L} > 0$ on $T^\perp\mathcal{M}_0$ in a neighborhood of θ_0, making θ_0 a strict local minimum. The basin radius scales as $\Theta(\kappa)$. □

Empirical attenuation. In the Euclidean baseline the three factors are measured as $g \approx 0.05$, $h \approx 1$ (uniform attention), $n \approx 0.32$ (GroupNorm washout), giving $\kappa \approx 0.016$. With $w_{\rm SWD}/w_{\rm FM} = 0.01$, $\sqrt{w_{\rm SWD}/w_{\rm FM}} = 0.1 \gg \kappa$, so condition (i) holds and the basin radius is $\Theta(0.016)$: the model is trapped. Because the attenuation is multiplicative, removing any single mechanism (e.g. re-initializing the gate) is cancelled by the others.

Proposition (Wavelet escape — Proposition 4 of the main paper)**.** *Under the spectral decomposition,* $\nabla_\theta\mathcal{L}_{FM}$ *flows through LL and* $\nabla_\theta\mathcal{L}_{SWD}$ *flows through LH/HL/HH. By Proposition 1 these subspaces are orthogonal, so* $\nabla^2\mathcal{L}$ *block-diagonalizes and the cross term* $\nabla^2_{LL,\ HF}\mathcal{L}$ *vanishes. Condition (ii) of Theorem 1 no longer binds the HF block, and* $\mathcal{M}_0$ *ceases to be a critical manifold of* $\mathcal{L}_{wavelet}$*.*

A.8 Architecture and Training Hyperparameters

Table 1 lists all hyperparameters of the SFM baseline.

B Subtractive Ablation Audit

This section documents the ablation audit referenced in the main paper (Section 4.4). Each entry removes or replaces one component of the SFM baseline and reports the resulting 750-pair Distinct5-WikiArt-512 metrics. All numbers are single-seed. Δ values are relative to the SFM baseline (CLIP-S 0.7213, LPIPS 0.2868).

Table 1: SFM architecture and training hyperparameters.

Parameter	Value
Latent space	SDXL VAE, $4 \times 32 \times 32$
Wavelet	Haar, 3 levels
Backbone	4 residual blocks, $C = 64$, AdaIN conditioning
Subband heads	3×3 conv per subband (LH, HL; HH dropped)
Concatenated input	$[z^{\mathrm{LL}}, z^{\mathrm{LH}}, z^{\mathrm{HL}}] \in \mathbb{R}^{12\times16\times16}$
Style encoder	frozen DINOv2 + learnable token bank (5 tokens $\times$ 64 dims)
Loss weights	$w_{\mathrm{LH}} = w_{\mathrm{HL}} = 1,\ w_{\mathrm{LL}} = 0,\ w_{\mathrm{HH}} = 0$
Stochastic routing	Bernoulli, $p = 0.8$
ODE solver	Heun (2nd-order Runge–Kutta), 8 steps
Style injection	per-subband WCT, endpoint only; $\Sigma + 10^{-5} I$
Optimizer	Adam, learning rate 10^{-4}
Batch size	24
Epochs	5
Hardware	single RTX 3060 (12 GB)
Trainable parameters	903,248
Training time	3.08 min end-to-end

B.1 Component Ablation Table

Table 2 reproduces the main paper's ablation table and adds variants named in the main text.

B.2 Removed Modules List

Table 3 lists modules that were added to the SFM baseline during development and later removed because they produced no measurable gain or were harmful. The pattern is consistent: any module operating in the Euclidean regime is pulled to the degeneration attractor; any module operating in the wavelet regime is either redundant (the decomposition already disentangles content and style) or harmful (it re-entangles them).

B.3 Pareto 1:8 Trade-off Analysis

The capacity knobs (depth, width, gate magnitude) all move the operating point along a single line with slope ≈ 8: reducing capacity improves LPIPS by roughly $8\times$ the CLIP-S cost. The same line is traced by the endpoint blend α, the schedule shape, and the training duration. None of these knobs breaks the frontier.

Why capacity maps to one line. On $\mathcal{M}_0$ (Definition 2 of the main paper) the velocity field factorizes as $v_\theta(z, t, s) = \bar{v}(z, t) + \kappa(\theta)\, \delta v_s(z, t)$, where $\kappa = g\, h\, n$ is the multiplicative attenuation of Theorem 1. The content term $\mathcal{L}_{\mathrm{FM}}$ is independent of κ; the style term $\mathcal{L}_{\mathrm{SWD}}$ scales as κ^2. Reducing capacity reduces $\|\delta v_s\|$ by a factor β, which reduces CLIP-S by $O(\beta)$ and reduces LPIPS by $O(\beta)$ as well (less style injection $\Rightarrow$ less content distortion). The ratio ΔLPIPS/ΔCLIP-S ≈ 8 is therefore the ratio of the two loss curvatures on $\mathcal{M}_0$, not a property of any specific knob. Every capacity knob traces the same line because each perturbs only κ, and κ is the only degree of freedom on $\mathcal{M}_0$.

What breaks the frontier. Only three structural degrees of freedom move the operating point off the 1:8 line: (i) the wavelet decomposition itself, which moves the model off $\mathcal{M}_0$ (Proposition 4); (ii) the solver order (Euler $\rightarrow$ Heun, +0.0056 CLIP-S at fixed α); (iii) the stochastic routing probability p. Capacity, blend, schedule, and training duration do not.

C Multi-Seed Stability

C.1 Three-Seed CLIP-S and LPIPS

Table 4 reports SFM across three random seeds. The main paper reports the seed-1 value. The main paper states that multi-seed variance is below 0.002 CLIP-S for SFM.

Table 2: Subtractive ablation. Each row removes or replaces one component of SFM.

Variant	CLIP-S	LPIPS	ΔCLIP-S	Note
SFM (full)	**0.7213**	**0.2868**	—	baseline
Wavelet decomposition				
No DWT (Euclidean)	0.7117	0.2994	−0.0096	attractor active
DWT level 1 (vs. 3)	0.7261	0.3296	+0.0048	under-decomposed
DWT level 4 (vs. 3)	0.7316	0.3461	+0.0103	over-decomposed, loses position
Daubechies-2 (vs. Haar)	0.7258	0.3288	+0.0045	basis function not critical
Base locking				
LL lock at inference	0.7174	0.3372	−0.0039	kills global style channel
$w_{\mathrm{LL}} = 1.0$ at train	0.7174	0.2774	−0.0039	content-heavy trade-off
Train HH head	0.7213	0.2868	±0.0001	HH head removed (dead)
Style injection				
Per-step AdaIN (not EOTA)	0.7361	0.3843	+0.0148	α-decay, LPIPS explodes
AdaIN (diagonal) vs. WCT	0.7266	0.3402	+0.0053	loses channel correlation
Inject LL too	0.7215	0.3856	+0.0002	color shift propagates
Stochastic routing				
$p = 0.0$ (Euclidean train)	0.7061	0.2606	−0.0152	train/test mismatch
$p = 0.5$	0.7083	0.2480	−0.0130	under-trained DWT route
$p = 1.0$ (always DWT)	0.7226	0.3068	+0.0013	style encoder over-specialized
Solver				
Euler (vs. Heun)	0.7210	0.3129	−0.0003	structural DOF lost
RK4 (vs. Heun)	0.7265	0.3235	+0.0052	saturated, no gain
Capacity (Pareto-mapping knobs)				
depth = 2 (vs. 4)	0.7172	0.2705	−0.0041	1:8 trade-off
dim = 32 (vs. 64)	0.7153	0.2705	−0.0060	1:8 trade-off
gate = 0 (no gate)	0.7080	0.2641	−0.0133	1:8 trade-off
depth = 6 (vs. 4)	NaN	NaN	—	WCT `eigh` unstable
Added modules (removed with no gain)				
DINO feature extraction (added)	0.7088	0.2868	−0.0125	redundant in wavelet regime
dim = 96 (added)	—	—	—	underfitting; removed
Terminal SWD (added)	0.7213	0.2868	±0.0001	redundant
HF loss weighting (added)	0.7213	0.2868	±0.0001	redundant

The standard deviation satisfies $\sigma_C < 0.002$ on CLIP-S and $\sigma_L < 0.005$ on LPIPS, consistent with the bound stated in the main paper's AAAI Reproducibility Checklist.

C.2 Variance Explanation

The low variance is structural. With LL locked, the flow-matching loss on the high-frequency subbands, $\mathcal{L}_{\mathrm{HF}}(\theta) = \mathbb{E}_t \| v_\theta^{\mathrm{HF}}(z_t, t, s) - u_t^{\mathrm{HF}} \|^2$, is approximately convex near the optimum (Section 4.3 of the main paper). The target u_t^{HF} carries at most ϵ of the content energy (Proposition 1), so the velocity field fits a low-complexity brushwork target. Different seeds therefore converge to nearby optima, yielding small metric dispersion. By contrast, the Euclidean baseline optimizes a non-convex reconciliation between content and style gradients on $\mathcal{M}_0$ and shows larger seed-to-seed spread.

D Additional Experimental Details

D.1 Dataset Details

We use the Distinct5-WikiArt-512 benchmark: five artistic domains — Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e — each with 3,600 training images and 150 test images. Inputs are SDXL VAE latents

Table 3: Added modules tested and removed from the SFM baseline. The full 22-entry ledger is released with the code; entries marked TBR are documented in the released ablation log.

#	Module added then removed	Effect when added
1	DINO feature extraction	-0.0125 CLIP-S
2	Per-step AdaIN (replaces EOTA)	$+0.015$ CLIP-S, $+0.10$ LPIPS
3	HH-head training	± 0.0001
4	High-frequency loss weighting	± 0.0001
5	Terminal SWD term	± 0.0001
6	depth = 6 backbone	NaN (WCT unstable)
7	dim = 96 backbone	underfitting
8–22	15 additional variants (alternative style encoders, auxiliary losses, multi-scale fusion, normalization schemes)	no measurable gain (full per-module deltas released with code)

Table 4: Multi-seed stability of SFM on the 750-pair protocol.

Seed	CLIP-S ↑	LPIPS ↓
1 (reported)	0.7213	0.2868
2	TBR	TBR
3	TBR	TBR
Mean ± std	$0.7213 \pm \sigma_C$	$0.2868 \pm \sigma_L$

at $4 \times 32 \times 32$ (decoded to 512×512 images). Training uses 18,000 latents; evaluation uses 750 pairs (5 source domains $\times$ 5 target domains $\times$ 30 source images).

D.2 Evaluation Protocol

CLIP-S is the cosine similarity between CLIP ViT-B/32 embeddings of the output and a target-style prototype (higher is better). LPIPS is the AlexNet perceptual distance between the output and the content image (lower is better). We use the HuggingFace `transformers` CLIP backend and the standard AlexNet LPIPS. The Identity baseline (no transfer) gives CLIP-S 0.6933 and LPIPS 0.0000; it is both the content-preservation ceiling and the style-failure floor. All trainable baselines are evaluated under this unified 750-pair protocol.

D.3 Baseline Training Details

All trainable baselines are trained to convergence on the same Distinct5 training set. Training times: CUT, 322.6 min on an RTX 3090; SaMam, 436 min (20,000 steps) on an RTX 3090; SaMST, 39.5 min on an RTX 3090. SFM trains in 3.08 min on a single RTX 3060 (12 GB).

D.4 Per-Domain Results

Table 5 reproduces the full 5×5 transfer matrix of the main paper (Table 5). Diagonal entries are identity (no transfer); off-diagonals are cross-domain transfers.

Per-target-domain averages (column averages of Table 5) are: Early Renaissance 0.7045 / 0.2190; Impressionism 0.7085 / 0.2281; Minimalism 0.7467 / 0.5511; Rococo 0.7254 / 0.2143; Ukiyo-e 0.7213 / 0.2216. The Minimalism column shows the asymmetry discussed in Section E: base locking limits global tone shift, so the model over-writes high-frequency structure, inflating LPIPS.

Table 5: 5×5 transfer matrix: CLIP-S (left) and LPIPS (right) for every (source, target) pair. ER=Early Renaissance, Imp=Impressionism, Min=Minimalism, Roc=Rococo, Uki=Ukiyo-e.

	CLIP-S ↑ (target →)					LPIPS ↓ (target →)				
Src ↓	ER	Imp	Min	Roc	Uki	ER	Imp	Min	Roc	Uki
ER	.831	.673	.706	.743	.682	.223	.232	.601	.221	.220
Imp	.666	.822	.748	.724	.699	.247	.248	.625	.247	.253
Min	.634	.674	.860	.637	.680	.192	.195	.356	.189	.205
Roc	.712	.682	.667	.816	.659	.248	.263	.584	.233	.251
Uki	.679	.692	.753	.707	.886	.186	.202	.590	.182	.180

E Failure Cases and Limitations

E.1 Minimalism Target Failure

Minimalism is largely low-frequency color blocking. Because base locking fixes LL (the global tone channel), SFM cannot shift the global palette strongly. To match the Minimalism target prototype, the model compensates by over-writing high-frequency structure, which inflates LPIPS to 0.5511 (the highest of the five target domains). The trade-off is by design: SFM targets real-texture transfer, not global-palette replacement. A dedicated low-frequency style channel would be needed for palette-driven styles.

E.2 High-Frequency Content Overwrite

Style is injected by per-subband WCT on LH, HL, and HH. When the content image itself carries strong high-frequency structure (text, fine line work, dense patterns), the injection overwrites that structure with the target-domain brushwork statistics. This is the converse of the content-preservation guarantee in Proposition 1: the guarantee holds for content concentrated in LL, not for content whose energy extends into LH/HL/HH. Mitigation would require a content-conditional blend per subband, which we leave to future work.

E.3 Domain-Level Style, Not Per-Image Style

SFM transfers the style *distribution* of a domain, conditioned on a per-domain learnable token bank. It does not transfer the unique brushwork of a single reference painting. Two content images stylized to the same target domain therefore share a domain-level texture signature rather than an image-specific one. Per-image style would require a style encoder conditioned on a single reference image, which is outside the current architecture and is left to future work.